

Pertuzumab – Drug Summary

Pertuzumab (brand name *Perjeta*, and co-formulated as *Phesgo* SC) is a humanized IgG1 monoclonal antibody targeting the extracellular domain of HER2/ERBB2. It is approved for **HER2-positive breast cancer** in combination with trastuzumab and chemotherapy. In the metastatic setting, pertuzumab plus trastuzumab and docetaxel improves progression-free and overall survival (median PFS ~18.5 vs 12.4 mo) ¹, establishing this dual blockade as first-line therapy for HER2+ disease. Pertuzumab is also approved in the neoadjuvant setting (with trastuzumab and chemo) to increase pathological complete response rates ², and as adjuvant therapy for high-risk early HER2+ breast cancer (the APHINITY trial showed improved invasive disease-free survival, especially in node-positive patients) ³. Its mechanism is complementary to trastuzumab: pertuzumab binds HER2's dimerization domain (subdomain II), preventing ligand-dependent HER2 heterodimerization and downstream signaling ⁴. This blocks key growth/survival pathways (MAPK, PI3K/AKT) and induces antibody-dependent cellular cytotoxicity (ADCC), leading to reduced proliferation and increased apoptosis of HER2-driven tumor cells ⁴ ⁵.

Identifiers & Synonyms

- **ChEMBL ID:** ChEMBL2007641 ⁶
- **DrugBank ID:** DB06366 ⁷
- **Synonyms / Development codes:** 2C4 Antibody; rhuMAb-2C4 (pertuzumab itself); BCD-178, MOAB-2C4, R1273, RG-1273, SYSA1901 ⁸ ⁹.
- **Trade/Brand names:** *Perjeta*, *Perjeta-Herceptin*, *Phesgo* (combination SC formulation with trastuzumab) ⁷.

Mechanism of Action in Breast Cancer

Pertuzumab is a HER2/neu (ERBB2) receptor antagonist. It binds to subdomain II of the HER2 ectodomain, which is essential for ligand-induced dimerization ⁴. By blocking dimerization, pertuzumab inhibits HER2-driven signal transduction. In particular, ligand-activated HER2 normally recruits PI3K and MAPK pathway effectors; pertuzumab binding prevents this, reducing downstream **PI3K/AKT** and **RAS/MAPK** signaling ⁴. The DrugBank summary explains: "Pertuzumab exerts its antineoplastic effects by binding to and inhibiting the activity of HER2... inhibition of HER2 prevents activation of MAPK and PI3K pathways" ⁴. Downstream, this leads to cell-cycle arrest and apoptosis in HER2-amplified tumor cells. Additionally, as an IgG1 antibody pertuzumab's Fc region engages immune effector cells; it mediates **ADCC (antibody-dependent cellular cytotoxicity)** against HER2+ tumor cells ⁴. In practice, pertuzumab is given with trastuzumab (which binds a different HER2 epitope) for dual HER2 blockade. Clinical trials (CLEOPATRA, NeoSphere, APHINITY, etc.) consistently show that combining pertuzumab with trastuzumab yields greater tumor response than trastuzumab alone ² ¹.

Primary Targets (HGNC)

- **ERBB2 (HGNC: ERBB2)** – HER2/neu. This receptor tyrosine kinase is the sole direct target of pertuzumab ⁴. By binding HER2's extracellular domain, pertuzumab inhibits HER2 activation and downstream signaling ⁴. No other human proteins are known to be directly bound by pertuzumab.

Pathways Overview

Pertuzumab's action perturbs multiple oncogenic pathways in HER2+ breast cancer. Key affected pathways include:

- **KEGG hsa04012 (ErbB signaling pathway)**: A major RTK cascade where ligand-bound HER-family receptors (EGFR/ERBB2/ERBB3/ERBB4) dimerize and activate MAPK and PI3K/AKT pathways. Pertuzumab blocks HER2 dimerization, thus **down-regulating ErbB signaling** ¹⁰ ⁴.
- **Reactome R-HSA-9006934 (Signaling by Receptor Tyrosine Kinases)**: RTKs (including HER2) typically dimerize and autophosphorylate, activating ERK/MAPK and AKT pathways ¹¹. Pertuzumab inhibits HER2-containing RTK complexes, leading to **reduced RTK cascade activation** ⁴ ¹¹.
- **KEGG hsa04151 (PI3K-Akt signaling)**: PI3K/AKT is activated by PIP3 upon RTK engagement and promotes growth/survival ¹². Pertuzumab (via HER2 blockade) **lowers PI3K/AKT signaling**, cutting pro-survival signals ⁴ ¹³.
- **Reactome R-HSA-1257604 (PIP3 activates AKT)**: This pathway describes PI3K-mediated AKT activation ¹³; pertuzumab's upstream inhibition reduces the flux through this pathway.
- **KEGG hsa04010 (MAPK signaling)**: The conserved MAPK cascade (ERK1/2, JNK, p38, etc) drives proliferation and differentiation ¹⁴. In HER2+ cells, HER2 contributes to MAPK activation. Pertuzumab **dampens MAPK signaling** by preventing HER2-mediated Raf/MEK/ERK activation ¹⁰ ¹⁴.
- **KEGG hsa04110 (Cell cycle)**: CDK/cyclin complexes govern cell cycle progression ¹⁵. With upstream mitogenic signals reduced by HER2 inhibition, cell-cycle progression is slowed (**down-regulated cell cycle**) ¹⁵. E2F-mediated S-phase entry is diminished.
- **KEGG hsa04210 (Apoptosis)**: The programmed cell death pathway is *up*-regulated. Loss of HER2 survival signaling shifts the balance toward apoptosis ¹⁶. Pertuzumab induces apoptotic pathways by removing prosurvival cues from the PI3K/AKT axis.
- **GO:0038093 (Fc receptor signaling pathway)**: An immune-effector pathway triggered by antibody Fc binding to Fcγ receptors ¹⁷. Pertuzumab's Fc engages NK and other immune cells to mediate ADCC, effectively **up-regulating this immune cytotoxicity** against HER2+ cells ⁴ ¹⁷.

Overall, pertuzumab is expected to **down-regulate** HER2-driven oncogenic pathways (RTK, ErbB, PI3K/AKT, MAPK, cell cycle) and **up-regulate** apoptotic and immune effector pathways. This broad pathway modulation aligns with its clinical activity in HER2-positive tumors ⁴ ¹⁷. (By contrast, up-regulation of PI3K/AKT or alternate RTKs would be expected to mediate *resistance* in treated cells.)

Breast Cancer Subtype–Stratified Evidence

Pertuzumab's benefit is confined to **HER2-positive** breast cancers. In patients whose tumors overexpress HER2 (either hormone-receptor-positive *HER2+* or triple-negative *HER2+* subtypes ¹⁸), pertuzumab plus

trastuzumab and chemo is a standard component of therapy ¹⁹ ¹ . For example, the CLEOPATRA trial in metastatic HER2+ disease (nearly equal HR+/HR- distribution ²⁰) showed improved outcomes with pertuzumab. In the neoadjuvant NeoSphere trial (HER2+ patients with varied HR status) adding pertuzumab raised pCR from 29.0% to 45.8% ² . In the APHINITY adjuvant trial, pertuzumab improved 8-year invasive DFS predominantly in node-positive HER2+ patients; benefit was seen in both HR- (ER-negative) and HR+ cohorts ²¹ . In summary, **all HER2+ subtypes** benefit from pertuzumab (Luminal B [HR+/HER2+], HER2-enriched [HR-/HER2+] alike ²¹).

In contrast, *HER2-negative* subtypes derive no benefit: triple-negative breast cancer (TNBC, HR-/HER2-) or hormone-positive/HER2- cancers are not indicated for pertuzumab. Regulatory guidance and trials only address HER2+ disease ¹⁹ ¹⁸ . Similarly, germline BRCA1/2 mutation status does not itself predict response; pertuzumab would only be used in BRCA-mutant cases that also have HER2 overexpression (if HER2-, PARP inhibitors or other therapies are appropriate instead).

Contraindications and Safety

Contraindications: The FDA label lists no absolute contraindications other than known hypersensitivity. The EMA specifies that pertuzumab is *contraindicated in patients with known hypersensitivity* to pertuzumab or excipients ²² . Pertuzumab should *not* be used during pregnancy or breastfeeding. It causes fetal harm in animal studies, and women of childbearing potential must use contraception ²³ . Embryo-fetal toxicity is a boxed warning in the label ²⁴ .

Major warnings/precautions: *Cardiac dysfunction:* HER2 blockade carries risk of left ventricular dysfunction. Pertuzumab can cause decreases in LVEF, especially in patients with prior anthracycline/radiation; LVEF must be assessed at baseline and periodically ²⁵ ²⁶ . If LVEF falls significantly (<40% or ≥10% drop), pertuzumab (and trastuzumab) should be withheld. *Infusion reactions:* Severe hypersensitivity and infusion reactions have been reported. Patients should be monitored during infusion and treated promptly if reaction occurs ²⁷ ²² . *Neutropenia/Febrile neutropenia:* Pertuzumab plus docetaxel increases febrile neutropenia risk (as seen in CLEOPATRA and other trials) ²⁸ ; prophylactic growth factors may be used per guidelines. *Other:* Diarrhea and mucositis are common AEs (typically low grade, but higher grade diarrhea occasionally occurs). In NeoSphere, the most frequent severe events were neutropenia, febrile neutropenia and leukopenia ²⁹ . There were no unexpected safety signals; cardiac and hematologic toxicity were similar to trastuzumab alone.

For pregnant or breastfeeding women, pertuzumab is not recommended (no clinical data, but animal studies show fetal toxicity) ²³ . Live vaccines should be avoided during therapy. Overall, tolerability is manageable: Grade 3–4 toxicities mostly relate to combination chemo (neutropenia, febrile neutropenia), and pertuzumab-specific effects (diarrhea, rash) are generally mild ²⁹ .

Clinical Trial and Guideline Context

Pertuzumab's clinical use stems from landmark trials:

- **CLEOPATRA (Baselga NEJM 2012):** First-line metastatic HER2+ breast cancer; pertuzumab+trastuzumab+docetaxel vs placebo arm. Median PFS was 18.5 vs 12.4 months (HR 0.62, $P<0.001$) ¹ , and later analysis showed significant overall survival benefit (19.5 month gain) with

pertuzumab. This established dual HER2 blockade as standard.

- **NeoSphere (Gianni Lancet Oncol 2012):** Neoadjuvant HER2+ early breast cancer; four arms including trastuzumab+docetaxel with or without pertuzumab. The pertuzumab arm achieved pCR in 45.8% of patients vs 29.0% without pertuzumab ². These results supported pertuzumab's use in early-stage HER2+ disease.

- **APHINITY (Loibl JCO 2024):** Adjuvant HER2+ disease (node-positive or high risk); adding pertuzumab to trastuzumab/chemo modestly improved invasive DFS. At 8.4 years median follow-up, 8-year invasive DFS in node+ patients was 86.1% with pertuzumab vs 81.2% without (HR 0.72) ²¹. Benefit was seen in both HR- and HR+ subgroups (HRs ~0.82 and 0.75) ³⁰, though overall absolute benefit was small. No OS improvement was seen at interim analysis ³¹.

- **PHEREXA (JCO 2017):** In patients who had progressed on first-line HER2 therapy, adding pertuzumab to trastuzumab+capecitabine gave a modest PFS benefit (8.2 vs 6.5 mo) but no significant OS gain. This niche finding has not changed practice significantly.

In practice, oncology guidelines (NCCN, ASCO, ESMO) now recommend pertuzumab as part of standard HER2-targeted regimens: *first-line metastatic* HER2+ patients receive trastuzumab + pertuzumab + taxane (Category 1), and *neoadjuvant/adjuvant* HER2+ candidates (especially node-positive) receive the same dual antibodies plus chemo. The FDA and EMA labels reflect these settings.

Additional Notes

- **Formulations:** Apart from intravenous infusions, pertuzumab is available in a fixed-dose subcutaneous combination (PHESGO) with trastuzumab and hyaluronidase for early HER2+ breast cancer patients who have tolerated IV therapy ⁷. This provides shorter administration.
- **Pharmacokinetics:** Pertuzumab has a prolonged half-life (~18 days, typical for human IgG1) with proteolytic catabolism; it is not metabolized by CYP enzymes. Steady-state levels are reached after ~8–9 infusions. Concurrent trastuzumab has no clinically significant PK interaction.
- **Immunology:** By virtue of its Fc, pertuzumab can trigger immune effectors. No routine biomarkers (other than HER2 expression) predict response. Potential resistance mechanisms include PIK3CA mutations or PTEN loss (activating PI3K/AKT independent of HER2) and alternate RTK signaling (e.g. IGF-1R, MET). Indeed, ~10–30% of breast cancers harbor PIK3CA mutations ³², which may attenuate response to HER2 blockade. No drugs specifically reverse pertuzumab resistance currently.
- **Adverse effects:** Besides cardiotoxicity and infusion reactions noted above, common adverse events include diarrhea, rash, mucositis, and electrolyte abnormalities. The safety profile is similar to trastuzumab alone; no new safety signals have emerged.

Pathway Evidence Summary

Pathway	Regulation (pertuzumab)	Effect	Rationale	References
ErbB (HER) signaling (KEGG hsa04012)	↓ (down)	↓ Tumor growth/survival signaling	Pertuzumab binds HER2 and blocks ligand-dependent HER2 dimerization ⁴ , shutting off downstream MAPK/PI3K pathways ¹⁰ .	⁴ ¹⁰
PI3K-AKT pathway (KEGG hsa04151; Reactome R-HSA-1257604)	↓ (down)	↓ Cell survival/metabolism signaling	PI3K is activated by phosphorylated HER2; pertuzumab prevents this activation ⁴ ¹³ . Akt phosphorylation and its pro-survival signals are reduced.	⁴ ¹³
MAPK pathway (KEGG hsa04010)	↓ (down)	↓ Proliferation/differentiation	HER2 blockade decreases Ras/Raf/ERK activation ¹⁰ ¹⁴ , leading to reduced mitogenic signaling.	¹⁴
Cell cycle (KEGG hsa04110)	↓ (down)	↓ Cell division	Decreased HER2/MAPK signaling leads to lower cyclin-CDK activity and E2F activation ¹⁵ , causing G1/S arrest.	¹⁵
Apoptosis (KEGG hsa04210)	↑ (up)	↑ Tumor cell apoptosis	Removal of HER2-driven survival signals shifts balance to apoptotic pathways ¹⁶ (caspase activation).	¹⁶
FcγR/ADCC signaling (GO: 0038093)	↑ (up)	↑ Immune-mediated tumor killing	Pertuzumab's Fc region engages Fcγ receptors on NK cells, triggering ADCC ⁴ ¹⁷ against HER2+ cells.	⁴ ¹⁷
Breast cancer (KEGG hsa05224)	↓ (HER2+ subtype genes)	Reduced HER2 oncogenic program	In HER2+ tumors, HER2 drives PI3K/AKT and MAPK signaling ³³ ; pertuzumab down-modulates this oncogenic program.	³³

Each pathway is supported by literature: pertuzumab directly inhibits HER2 (ERBB2) activation ⁴, which in turn down-regulates PI3K/AKT and MAPK signaling ¹² ¹³, leading to cell-cycle arrest ¹⁵ and apoptosis ¹⁶. Simultaneously, pertuzumab's Fc activates immune effector pathways ¹⁷. These regulatory effects are consistent with its clinical efficacy in HER2-positive breast cancer.

Sources: Authoritative data are drawn from FDA/EMA labeling, NCCN/ASCO reviews, PubMed publications, ChEMBL/DrugBank databases, and pathway resources (KEGG/Reactome/GO) as indicated by citations. All data pertain to human breast cancer (Homo sapiens) and approved indications.

¹ Pertuzumab plus trastuzumab plus docetaxel for metastatic breast cancer - PubMed

<https://pubmed.ncbi.nlm.nih.gov/22149875/>

² ⁵ ²⁹ Efficacy and safety of neoadjuvant pertuzumab and trastuzumab in women with locally advanced, inflammatory, or early HER2-positive breast cancer (NeoSphere): a randomised multicentre, open-label, phase 2 trial - PubMed

<https://pubmed.ncbi.nlm.nih.gov/22153890/>

³ ²¹ ³⁰ ³¹ Adjuvant Pertuzumab and Trastuzumab in Early Human Epidermal Growth Factor Receptor 2-Positive Breast Cancer in the APHINITY Trial: Third Interim Overall Survival Analysis With Efficacy Update - PubMed

<https://pubmed.ncbi.nlm.nih.gov/39259927/>

⁴ ⁶ ⁷ ⁸ ⁹ Pertuzumab: Uses, Interactions, Mechanism of Action | DrugBank Online

<https://go.drugbank.com/drugs/DB06366>

¹⁰ KEGG PATHWAY: hsa04012

https://www.genome.jp/dbget-bin/www_bget?pathway:hsa04012

¹¹ Reactome | Signaling by Receptor Tyrosine Kinases

<https://reactome.org/content/detail/R-HSA-9006934>

¹² KEGG PATHWAY: PI3K-Akt signaling pathway - Homo sapiens (human)

<https://www.kegg.jp/pathway/hsa04151>

¹³ Reactome | PIP3 activates AKT signaling

<https://reactome.org/content/detail/R-HSA-1257604>

¹⁴ KEGG PATHWAY: MAPK signaling pathway - Homo sapiens (human)

<https://www.kegg.jp/pathway/hsa04010>

¹⁵ KEGG PATHWAY: Cell cycle - Homo sapiens (human)

<https://www.kegg.jp/pathway/hsa04110>

¹⁶ KEGG PATHWAY: Apoptosis - Homo sapiens (human)

<https://www.kegg.jp/pathway/hsa04210>

¹⁷ OakMine PM1N: GOterm GO:0038093 Fc receptor signaling pathway GO

https://urgi.versailles.inrae.fr/OakMine_PM1N/report.do?id=6431389

¹⁸ ³² ³³ KEGG PATHWAY: hsa05224

https://www.genome.jp/dbget-bin/www_bget?pathway:hsa05224

19 20 24 26 27 **Perjeta (Pertuzumab) Label**

https://www.accessdata.fda.gov/drugsatfda_docs/label/2012/125409lbl.pdf

22 23 25 28 **Perjeta, INN-pertuzumab**

https://ec.europa.eu/health/documents/community-register/2016/20161027136303/anx_136303_en.pdf